

Information Collection, Classification and Knowledge Base Development report

1. Introduction: Information Collection Report

This report outlines the systematic approach for executing Task 3: Information Collection, which aims to collect relevant and verified information about Mfano Bora Africa to power the chatbot.

Using the groundwork laid in Task 2, we will gather factual data from identified primary sources. These sources include the Mfano Bora official web portals (such as the careers and attachment application pages), institutional records detailing initiatives like the Road Safety Club Kenya, operational registries containing contact metadata, and curated administrative FAQs. By extracting data directly from these verified avenues, we guarantee that the resulting system relies on a definitive single source of truth, thereby eliminating AI hallucinations and providing high-quality, actionable responses to users.

2. Automated Web Scraper & Supabase Data Pipeline

To operationalise the collection process, an automated data ingestion pipeline will be developed. This pipeline will programmatically extract, process, and store the information in a Supabase PostgreSQL database.

The Data Workflow

1. Extraction (Web Scraping): Python-based scrapers (utilising libraries like BeautifulSoup or Scrapy) will automatically navigate the Mfano Bora website and related digital portals to extract raw text.
2. Text Cleaning & Chunking: The raw scraped text will be sanitised to remove duplicate HTML tags and outdated information, then broken into structured, manageable text chunks.
3. Vectorisation: Using open-source Hugging Face Sentence Transformers (e.g., `gte-small`), these text chunks will be converted into numerical vector embeddings.
4. Database Storage (Supabase): The processed chunks and their corresponding vector embeddings will be saved directly into the Supabase PostgreSQL database using the `pgvector` extension.

Database Schema

Following the defined file structure, the schema will be initialised via the `database/init\_schema.sql` script. The schema maps directly to the chatbot's requirements for semantic search:

```
```sql
-- Enable the pgvector extension in Supabase
CREATE EXTENSION IF NOT EXISTS vector;

-- Create the structured knowledge base table
CREATE TABLE knowledge_base (
 id BIGSERIAL PRIMARY KEY,
 source_url TEXT NOT NULL,
 category TEXT NOT NULL,
 content TEXT NOT NULL,
 embedding VECTOR(384) -- Dimension matches the Hugging Face embedding model
);
```
```

3. Evaluating Coverage of Dependent Tasks

Implementing this automated scraping and vector storage procedure naturally fulfils the requirements of subsequent assignment tasks:

Covers TASK 4: Information Classification? Yes

Requirement: Classify the collected information into appropriate information categories for chatbot retrieval.

Justification: During the extraction and chunking phase, the Python scripts will dynamically assign metadata tags to each text chunk based on the source URL and topic (e.g., tagging a chunk scraped from `/apply-now/` as "Careers & Attachments"). When this data is inserted into the `knowledge\_base` table under the `category` column, the information is effectively and systematically classified.

Covers TASK 5: Knowledge Base Development? Yes

Requirement: Design and develop a structured knowledge base containing the verified information required by the chatbot.

Justification: The creation of the PostgreSQL table using the `pgvector` extension directly serves as the structural foundation of the knowledge base. By securely storing the cleaned text chunks alongside their vector representations, this database acts as the exact structured environment required by the Retrieval-Augmented Generation (RAG) engine to conduct similarity searches and generate accurate Groq API responses.

4. Integration with Project File Structure

This automated collection plan aligns seamlessly with the established modular file architecture:

`ai-pipeline/data-cleaning/`: Will house the Python web scraping scripts responsible for extracting and sanitising the Mfano Bora website data.

`ai-pipeline/embeddings/`: Will contain the Hugging Face vectorisation scripts that convert the text into numerical models.

`ai-pipeline/knowledge-base/`: Will act as the staging directory for any raw, verified JSON/CSV data files before they are pushed to the database.

`database/init\_schema.sql`: Will contain the SQL commands to configure the Supabase `pgvector` tables.